Research Proposal: Metacognitive Medical Digital Twins

Bridging the Socio-Technical Gap via Multi-Reward GRPO Alignment

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1 Statement of the Problem

Current Clinical Decision Support Systems (CDSS) and Large Language Models (LLMs) in healthcare suffer from a critical Socio-Technical Gap whch is a disconnect between high-precision predictive modeling and the communicative needs of clinicians and patients. This gap manifests in three fundamental challenges that compromise both clinical effectiveness and patient safety.

First, existing systems exhibit Context Decay, treating longitudinal ICU data as static snapshots rather than dynamic biological processes. Recent critical physiological signals are inadequately prioritized over stale records, leading to physiological hallucinations which are the predictions that statistically plausible yet biologically impossible, such as suggesting a heart rate of 200 bpm for a sedated patient.

Second, current systems demonstrate Black-Box Reasoning, lacking auditable logic during high-stakes clinical transitions, particularly during the onset of Sepsis or Acute Kidney Injury (AKI). Clinicians cannot comprehend the causal pathways underlying AI recommendations, which erodes clinical trust and constrains adoption in safety-critical scenarios where accountability is paramount.

Third, architectures suffer from an Empathy Deficit, lacking Theory of Mind (ToM) mechanisms that would enable them to adapt communication to diverse cognitive and literacy profiles across patient populations. This failure to calibrate medical communication to varying health literacy levels precipitates poor health outcomes, elevated patient anxiety, and exacerbates health disparities particularly affecting vulnerable and medically underserved communities.

This project develops a Metacognitive Medical Digital Twin (MDT) which is a self-correcting, empathetic, and biologically grounded clinical agent. Using Group Relative Policy Optimization (GRPO), the aim is to align the LLM model like MedGemma-4B foundation model with four specific pillars: (1) Deductive Integrity to be transparent, step-by-step reasoning via structured cognitive streams; (2) Mechanistic Realism to have predictions grounded in standardized LOINC physiological trajectories; (3) Structural Empathy to model patient belief states through Theory of Mind; and (4) Biological safety which can enforce physiological boundary constraints via reinforcement learning.

2 Tasks to Accomplish

1. **Logical Bootstrapping via Supervised Fine-Tuning.** Establish a "Cognitive Scratchpad" architecture through domain-specific supervised fine-tuning, partitioning the model's internal reasoning into three auditable XML streams: <think> for deductive clinical logic, <patient_state> for physiological state estimation, and <user_belief> for empathetic user modeling. This foundational structure enables transparent clinical reasoning, establishing Deductive Integrity.

2. **Socio-Cognitive Engineering and Theory of Mind Implementation.** Integrate an ex-

plicit Theory of Mind module that dynamically assesses user health literacy levels and emotional states. The system maps patient responses to validated health literacy scales and adjusts communication complexity, terminology, and affective tone in real-time, operationalizing Structural Empathy.

3. **Mechanistic Realism via Temporal Physiological Simulation.** Develop a temporal grounding module that transforms discrete Electronic Health Record observations into continuous physiological trajectory predictions mapped to standardized LOINC codes. The system forecasts multi-step temporal evolution of critical biomarkers, implementing recency-weighted attention mechanisms to establish Mechanistic Realism.

4. **Multi-Reward Policy Alignment via GRPO.** Deploy a comprehensive Group Relative Policy Optimization framework using a modular training pipeline that simultaneously optimizes clinical accuracy, proactive surge detection, biological safety constraints, empathetic communication, and metacognitive depth. This phase leverages parameter-efficient fine-tuning through Low-Rank Adaptation with quantization, enforcing Biological Safety.

3 Datasets

Medical-O1-Reasoning-SFT Dataset. Provides Chain-of-Thought reasoning trajectories for complex medical problem-solving, encompassing differential diagnosis, multi-step deductive reasoning, and explicit self-correction protocols. Each instance contains expert-annotated reasoning chains supporting the development of Deductive Integrity.

MIMIC-IV Critical Care Database. Extracts longitudinal time-series trajectories for critical physiological markers including Heart Rate, Systolic Blood Pressure, Oxygen Saturation, Serum Lactate, and Serum Creatinine from ICU patient admissions. Data is preprocessed into standardized prediction windows with physiologically-bounded outlier removal, grounding Mechanistic Realism.

Standardized Clinical Ontologies (SNOMED-CT & LOINC). All clinical entities are mapped to universally recognized terminologies such as SNOMED Clinical Terms for diagnoses and procedures, LOINC for laboratory results and vital signs to ensure Electronic Health Record interoperability and clinical actionability.

4 Approach/Method

Triple-Stream Cognitive Architecture. The model's reasoning is partitioned into three XML-formatted streams enabling auditable decision-making: <think> captures deductive clinical reasoning and differential diagnosis (Deductive Integrity); <patient_state> maintains real-time physiological state representation grounded in LOINC-coded biomarkers (Mechanistic Realism); <user_belief> models patient health literacy, emotional state, and informational needs via Theory of Mind inference (Structural Empathy). This architectural separation ensures independent evaluation and optimization of each clinical reasoning dimension.

Multi-Reward Engine. The MDT is optimized through a custom total reward function that balances five clinical objectives as in the following equation:

$$R_{\text{total}} = w_1 R_{\text{sem}} + w_2 R_{\text{meta}} + w_3 R_{\text{emp}} + w_4 R_{\text{physio}} + w_5 R_{\text{bound}} \tag{1}$$

where R_{meta} represents the **Delta-Embedding Reward**, calculating cosine similarity shifts in latent space to verify genuine metacognition and penalize reward hacking. The five reward components are:

1. **Semantic Fidelity (R_{sem}):** Measures accuracy of clinical responses against expert-annotated discharge summaries using ROUGE-L and BERTScore metrics.

2. **Metacognitive Depth ($R_{\textbf{meta}}$):** Computes cosine similarity shifts in latent reasoning states to detect genuine self-correction, distinguishing substantive refinements from superficial rewording.
3. **Structural Empathy ($R_{\textbf{emp}}$):** Evaluates communication calibration to user profiles through readability indices and sentiment matching across health literacy levels.
4. **Proactive Surge Prediction ($R_{\textbf{physio}}$):** Rewards anticipation of physiological deterioration before ground truth occurrence, enabling early clinical intervention.
5. **Biological Safety Constraints ($R_{\textbf{bound}}$):** Penalizes predictions violating clinical reference ranges or suggesting biologically impossible vital sign combinations.

The weight parameters $(w_1, w_2, w_3, w_4, w_5)$ are empirically tuned to balance competing objectives, prioritizing clinical accuracy and patient safety while maintaining metacognitive transparency and empathetic communication.

Parameter-Efficient Implementation. The training strategy employs Low-Rank Adaptation (LoRA) with quantization to enable computationally feasible fine-tuning of the MedGemma-4B foundation model. LoRA targets specific attention and feed-forward layers while freezing the majority of base model parameters. The GRPO algorithm operates on batches of multiple generations per prompt, computing group-relative advantages to reduce variance and improve sample efficiency. Training is distributed across multiple high-performance GPUs with gradient accumulation to achieve effective large batch sizes while maintaining memory feasibility.

5 Experiment and Evaluation Plan

The validation framework employs comprehensive multi-dimensional benchmarking to assess clinical readiness across the four established pillars. Physiological trajectory accuracy is evaluated by measuring prediction error metrics for vital signs and laboratory values over standardized time horizons, with surge detection performance assessed through sensitivity and specificity analysis for clinically significant deterioration events, validating Mechanistic Realism and early warning capacity.

Deductive Integrity is verified through automated structural analysis of the three cognitive streams, ensuring proper formatting and valid medical logic. Board-certified intensivists conduct blinded expert reviews, rating the quality of differential diagnoses, evidence chains, and self-correction episodes using validated clinical competency frameworks.

Structural Empathy is assessed through human evaluation studies spanning patient profiles that vary across health literacy levels and emotional states. Clinical experts and patient advocates evaluate whether explanations are appropriately calibrated to inferred user characteristics, measuring language complexity through validated readability metrics compared against health literacy best practices.

Biological Safety is validated by quantifying physiological hallucination frequency through identification of predictions violating established clinical reference ranges. Systematic edge case testing with extreme vital signs and impossible laboratory value combinations validates enforcement of boundary constraints.

Finally, ablation studies establish individual reward component contributions through systematic removal and performance degradation measurement across all evaluation metrics. Rigorous statistical testing demonstrates that each component addresses a distinct Socio-Technical Gap dimension, providing empirical evidence that the complete multi-reward framework is necessary for comprehensive clinical performance.